

Caculus

Differentiation

Derivative: $\frac{d}{dx} f(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$

Higher-Order Derivatives: $\frac{d^n}{dx^n} f(x)$

Taylor's Formula: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$

Integration

Definite Integral: $\int_a^b f(x) dx = F(b) - F(a)$ where $F'(x) = f(x)$

Indefinite Integral: $\int f(x) dx = F(x) + C$ where $F'(x) = f(x)$

Matrix Calculus

scalar-by-vector Derivative: $\frac{\partial y}{\partial \mathbf{x}} = \left[\frac{\partial y}{\partial x_1} \quad \frac{\partial y}{\partial x_2} \quad \dots \quad \frac{\partial y}{\partial x_n} \right]$

vector-by-scalar Derivative: $\frac{\partial \mathbf{y}}{\partial x} = \begin{bmatrix} \frac{\partial y_1}{\partial x} \\ \frac{\partial y_2}{\partial x} \\ \vdots \\ \frac{\partial y_m}{\partial x} \end{bmatrix}$

vector-by-vector Derivative (Jacobian): $\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \dots & \frac{\partial y_1}{\partial x_n} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \dots & \frac{\partial y_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_m}{\partial x_1} & \frac{\partial y_m}{\partial x_2} & \dots & \frac{\partial y_m}{\partial x_n} \end{bmatrix}$

scalar-by-vector Second Derivative (Hessian): $\frac{\partial^2 y}{\partial \mathbf{x}^2} = \begin{bmatrix} \frac{\partial^2 y}{\partial x_1^2} & \frac{\partial^2 y}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 y}{\partial x_1 \partial x_n} \\ \frac{\partial^2 y}{\partial x_2 \partial x_1} & \frac{\partial^2 y}{\partial x_2^2} & \dots & \frac{\partial^2 y}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 y}{\partial x_n \partial x_1} & \frac{\partial^2 y}{\partial x_n \partial x_2} & \dots & \frac{\partial^2 y}{\partial x_n^2} \end{bmatrix}$

Chain Rule: If $\mathbf{y} = f(\mathbf{u})$ and $\mathbf{u} = g(\mathbf{x})$, then $\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \frac{\partial \mathbf{y}}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial \mathbf{x}}$

Common Derivatives

- $\frac{d}{dx}(c) = 0$
- $\frac{d}{dx}(x^n) = nx^{n-1}$
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(\ln x) = \frac{1}{x}$
- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$
- $\frac{d}{dx}(uv) = u'v + uv'$
- $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$
- $\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$
- $\frac{d}{dx}\left(\int_a^x f(t) dt\right) = f(x)$
- $\frac{d}{dx}\left(\int_x^b f(t) dt\right) = -f(x)$
- $\frac{d}{dx}\left(\int_{u(x)}^{v(x)} f(t) dt\right) = f(v(x))v'(x) - f(u(x))u'(x)$
- $\frac{d}{dx}(\mathbf{a}^T \mathbf{x}) = \mathbf{a}$
- $\frac{d}{dx}(\mathbf{x}^T \mathbf{A} \mathbf{x}) = (\mathbf{A} + \mathbf{A}^T) \mathbf{x}$
- $\frac{d}{dx}(\|\mathbf{x}\|_2^2) = 2\mathbf{x}$
- $\frac{d}{dx}(\|\mathbf{x}\|_1) = \text{sign}(\mathbf{x})$ where $\text{sign}(x_i) = \begin{cases} 1 & x_i > 0 \\ -1 & x_i < 0 \\ 0 & x_i = 0 \end{cases}$
- $\frac{d}{dx}(\det(\mathbf{A})) = \det(\mathbf{A}) \cdot \text{tr}(\mathbf{A}^{-1} \frac{d\mathbf{A}}{dx})$
- $\frac{d}{dx}(\text{tr}(\mathbf{A})) = \text{tr}(\frac{d\mathbf{A}}{dx})$
- $\frac{d}{dx}(\mathbf{A}^{-1}) = -\mathbf{A}^{-1} \frac{d\mathbf{A}}{dx} \mathbf{A}^{-1}$
- $\frac{d}{dx}(\mathbf{A}\mathbf{B}) = \frac{d\mathbf{A}}{dx} \mathbf{B} + \mathbf{A} \frac{d\mathbf{B}}{dx}$
- $\frac{d}{dx}(\mathbf{A} + \mathbf{B}) = \frac{d\mathbf{A}}{dx} + \frac{d\mathbf{B}}{dx}$
- $\frac{d}{dx}(\mathbf{A} \circ \mathbf{B}) = \frac{d\mathbf{A}}{dx} \circ \mathbf{B} + \mathbf{A} \circ \frac{d\mathbf{B}}{dx}$
- $\frac{d}{dx}(\mathbf{A} \otimes \mathbf{B}) = \frac{d\mathbf{A}}{dx} \otimes \mathbf{B} + \mathbf{A} \otimes \frac{d\mathbf{B}}{dx}$
- $\frac{d}{dx}(\|\mathbf{A}\|_F^2) = 2\mathbf{A}$